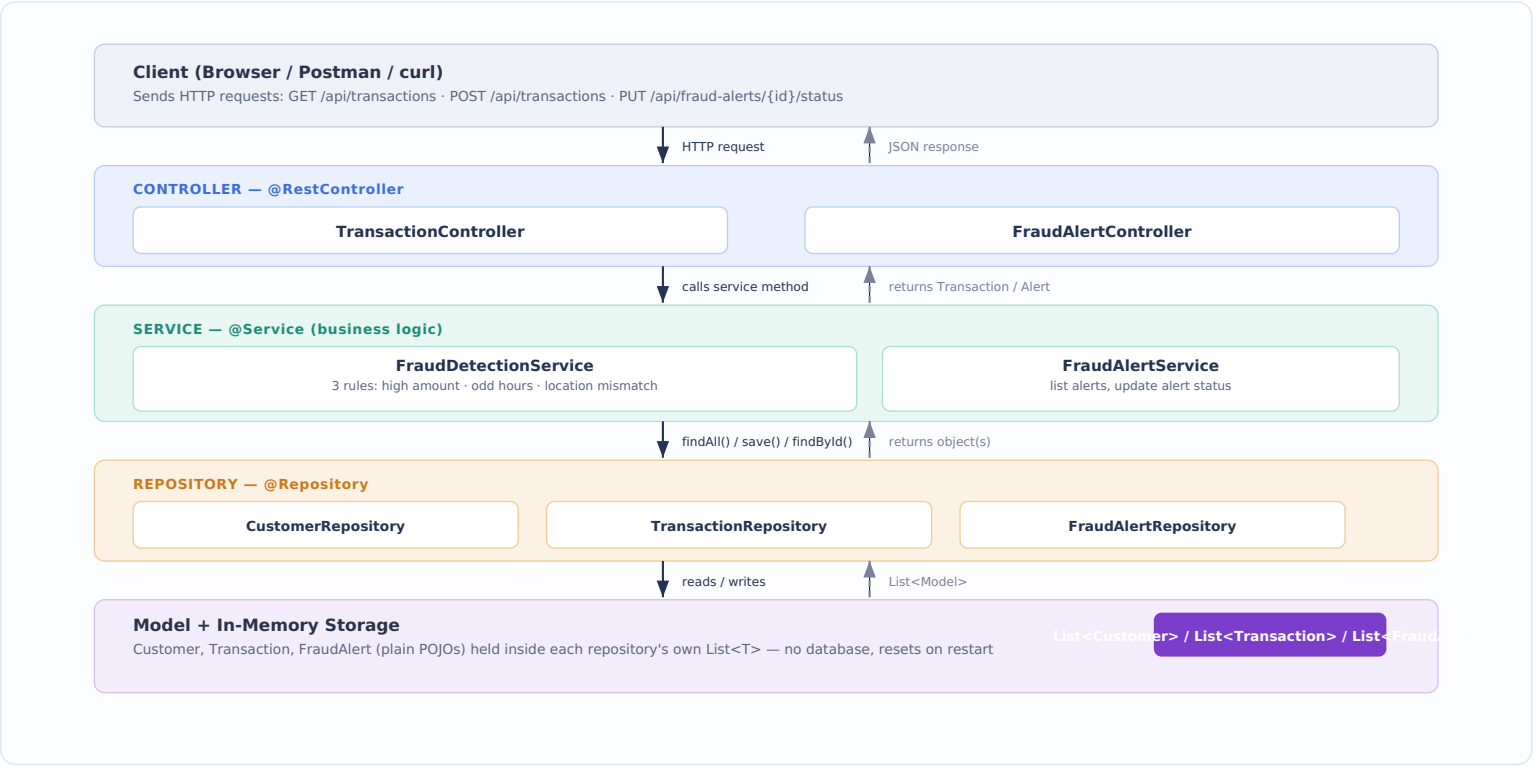


Fraud Detection System — Component Interactivity Diagram

The current version of the app: three layers plus an in-memory model, no database. Each request flows straight down through Controller → Service → Repository and back up — the exact same shape we'll reuse later when the Repository is swapped for JDBC + MySQL.



LAYER	CLASSES	WHAT IT DOES HERE
Controller	TransactionController, FraudAlertController	Receives HTTP requests, reads the request body, calls exactly one service method, returns the result as JSON.
Service	FraudDetectionService, FraudAlertService	All the actual decision-making lives here — runs the 3 fraud rules, calculates risk score, decides PROCESSED vs FLAGGED.
Repository	CustomerRepository, TransactionRepository, FraudAlertRepository	Owns one List<T> each. Provides findAll(), findById(), save() — same method names we'll reuse when this becomes JDBC later.
Model	Customer, Transaction, FraudAlert	Plain Java objects (POJOs) with fields + getters/setters. No annotations, no database mapping — just data.